Testing Guide

The product fails in four expensive ways: one tenant sees another's data, a bill is wrong, a payment and an order disagree, or the kitchen misses an order. Testing effort is weighted toward those four, with lighter coverage elsewhere.

1. Test layers

Layer	Tool	What it covers	Runs
Static	ESLint, `tsc --noEmit`	Types, lint rules	Every commit
Unit	Vitest	`computeBill`, state machine, Zod schemas, formatters	Every commit
Database policy	pgTAP via `supabase test db`	RLS for every table and role	Every commit
API integration	Vitest against local Supabase	Route handlers: auth, validation, idempotency, webhooks	Every PR
Component	Vitest + Testing Library	Item sheet rules, cart, steppers, KDS ticket	Every PR
End-to-end	Playwright (mobile and desktop projects)	Full journeys across guest, kitchen, dashboard	Every PR on preview
Accessibility	axe via Playwright	Guest and dashboard screens	Every PR
Visual regression	Playwright screenshots	Primitives and key screens	Every PR
Load	k6	Order placement and realtime fan-out	Before each release
Security	Manual checklist plus automated probes	Section 6	Before each release
Manual exploratory	Real devices	Section 9	Before pilot and each release

2. Coverage targets

Area	Target
`packages/core` (billing, state machine)	100% branches
Route handlers	90% lines; every handler has an unauthorised-path test
RLS	Every table has select, insert, update, delete tests per role
UI components	Behavioural tests for anything with rules; no snapshot-only tests
End-to-end	Every P0 requirement in the PRD maps to at least one spec

3. Test data

- `supabase/seed.sql` creates two tenants (A and B), each with one dine-in outlet and one canteen outlet, 5 tables, 4 categories, 30 items with variants and add-ons, and one user per role.
- Tests create their own orders and clean up, or run inside a transaction.
- Payments use the gateway's test mode and a local webhook simulator that signs payloads with the test secret.
- Never use real customer data or live keys in tests.

4. Critical unit cases

computeBill

Case	Expectation
Single item, 5% tax	Tax split evenly into CGST and SGST; total = subtotal + tax
Variant delta and two add-ons, qty 3	Line total = (price + delta + add-ons) x 3
Mixed tax rates across items	Tax computed per line, then summed
Service charge on and off	Applied on subtotal per outlet config

Case	Expectation
Rounding	Integer paise throughout; final rounding rule applied once and reported
Zero-price item, empty cart	Valid zero line; empty cart rejected
Large quantities	No overflow; no float drift
Client total differs from server total	Server value wins; mismatch returned as a specific error code

Order state machine

Case	Expectation
Every allowed transition per role	Succeeds and writes one `order_events` row
Every disallowed transition	Rejected with `invalid_transition`
Kitchen attempts reject or refund	Forbidden
Guest cancels after `accepted`	Rejected
Canteen order accepted before payment captured	Rejected
Same transition sent twice	Second call is a no-op, no duplicate event

5. API integration cases

Area	Cases
Guest session	Valid QR opens a session; unknown, inactive or regenerated token returns 404; suspended tenant returns the closed page
Place order	Happy path; sold-out item rejected; add-on min/max enforced; tampered price ignored; missing idempotency key rejected; repeated key returns the original order
Order visibility	A guest session cannot read another session's order by guessing ids

Area	Cases
Status change	Role checks; cross-outlet attempt forbidden
Menu	Availability toggle revalidates the cached menu
Payments	Amount derived from DB; signature verification failure rejected; partial payment leaves balance due
Webhook	Bad signature 400; duplicate delivery is idempotent; `captured` arriving before client verify still settles; `failed` leaves order unpaid; refund event updates payment
Rate limits	Burst of order placements from one session is throttled

6. Security and isolation tests

Test	Pass condition
Tenant A staff selects tenant B rows, every table	Zero rows
Tenant A staff inserts or updates with tenant B ids	Denied
Anonymous client queries any table directly	Denied
Manager scoped to outlet 1 reads outlet 2	Denied
Kitchen role calls refund, staff or settings endpoints	403
Service role key searched for in client bundles	Not present
IDOR sweep on all `/api/*/[id]` routes	403 or 404 for foreign ids
XSS payloads in item names, notes, guest names	Rendered as text on guest, KDS and dashboard
Oversized or wrong-type image upload	Rejected
PIN brute force	Locked out after configured attempts
Webhook replay with old timestamp	Rejected or harmless

7. End-to-end scenarios

Each runs with multiple browser contexts in one Playwright test: guest phone, kitchen tablet, manager desktop.

ID	Scenario	Key assertions
E2E-01	Dine-in happy path: scan, add 3 items with a variant and add-on, place, kitchen starts and readies, waiter serves, guest pays by UPI test, manager closes table	Each status reaches the guest in under 2 s; bill equals `computeBill`; table returns to free; invoice generated
E2E-02	Second order on the same table	Both orders on one running bill
E2E-03	Canteen happy path: scan counter QR, order, prepay, token shown, ready alert, handover	Unpaid order never reaches the kitchen; token appears then clears on pickup display
E2E-04	Item sold out while in cart	Guest sees a clear message and can remove and continue
E2E-05	Manager rejects an order with a reason	Guest sees the reason; paid amount refunded in test mode
E2E-06	Guest cancels while `placed`; cannot after `accepted`	Cancel button state matches
E2E-07	Payment fails then succeeds on retry	One captured payment; no duplicate charge
E2E-08	Pay at counter with cash	Waiter records cash; session closes
E2E-09	Kitchen screen goes offline for 30 s during two new orders	Banner shown; on reconnect both tickets present, in order
E2E-10	Guest refreshes during cart and during tracking	Cart and tracking state restored
E2E-11	Double tap on Place order	Exactly one order
E2E-12	QR regenerated	Old link shows an invalid-code page; new link works
E2E-13	Owner onboarding: sign up, create outlet, build menu, create tables, download QR PDF	Completed with no errors; PDF has one page per table

ID	Scenario	Key assertions
E2E-14	Call waiter and request bill	Appears on the dashboard in realtime and can be acknowledged
E2E-15	Analytics after seeded day	Totals match hand-computed values

8. Non-functional tests

Area	Method	Gate
Guest performance	Lighthouse CI, mobile profile, on menu and tracking pages	Performance 90+, LCP under 2.5 s, CLS under 0.1
Bundle size	Build analysis	Guest route JS under 200 KB gzipped
Load	k6: 50 outlets x 50 concurrent orders over 10 min	p95 place-order under 800 ms; error rate under 0.5%; no lost realtime events
Realtime latency	Scripted status change with timestamp comparison	p95 under 2 s
Accessibility	axe automated plus keyboard and screen-reader pass on guest flow	Zero serious or critical issues
Resilience	Throttle to slow 3G and offline in Playwright	Graceful messages; no data loss in cart

9. Manual device checklist

- Scan a printed QR with the default camera on a low-end Android phone and an iPhone; menu opens in the default browser and in in-app browsers.
- UPI intent flow hands off to a UPI app and returns to the tracking page.
- KDS on a tablet for a 2-hour soak: screen stays awake, sound alerts work after idle, memory stable.
- Dashboard on a 1366 x 768 laptop and a tablet.
- Printed QR readable at 30 cm under dim light.
- Text at 200% zoom on the guest app.

10. CI pipeline and release gates

Stage	Steps	Blocks merge
1. Static	Lint, typecheck	Yes
2. Unit and policy	Vitest, pgTAP	Yes
3. Integration	API tests on local Supabase	Yes
4. Preview	Deploy preview, Playwright, axe, Lighthouse	Yes
5. Release	k6 load, security checklist, manual device checklist	Blocks release

A release ships only when stages 1 to 5 pass, no open severity-1 or severity-2 bugs exist, and migrations have been rehearsed on a copy of production.

11. Bug severity

Severity	Definition	Response
S1	Data leak across tenants, wrong charge, orders lost	Fix immediately; block release
S2	Core flow broken for a role with no workaround	Fix before release
S3	Degraded flow with a workaround	Next release
S4	Cosmetic	Backlog

12. Monitoring after launch

Tests continue in production as monitors: a synthetic guest order on a hidden test outlet every 10 minutes, alerts on webhook failures, payment success rate below 90%, realtime disconnect spikes, and any order stuck in `placed` for more than 5 minutes during open hours.